

SVKM's NMIMS Deemed-to-be University
Mukesh Patel School of Technology Management and Engineering

Program: B Tech, MBA Tech & B Tech Integrated Data Science				Semester: V/ IX	
Course: Introduction to Modern Application Development				Code: 702DB0C018	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Android App Development					
Course Objective The aim of the course is to provide students with the tools and technologies to build, deploy, test, run, and manage full stack applications. It will empower students to deal with both the Frontend (client-side) and Backend (server-side) of a tech product. It will educate students to position them as full stack developers, proficient in several technologies enabling them to develop robust tech products, and thus play a major role in decision making for the organization.					
Course Outcomes After completion of the course, students will be able to - 1. Describe modern front-end technologies by outlining advanced HTML/CSS, Bootstrap /Tailwind utilities, and JavaScript ES6+ features with async patterns. (L2 – Understanding) 2. Apply Python and Django –covering models, views, templates, and REST APIs –to construct functional back-end services and integrate machine-learning models. (L3 – Applying) 3. Design data-driven applications that use PostgreSQL and MongoDB, and deploy them on cloud IaaS platforms such as AWS S3 or DigitalOcean. (L4 – Analyzing) 4. Develop interactive single-page applications in React JS by implementing JSX, hooks, the Context API, and secure API connectivity. (L3 – Applying) 5. Implement end-to-end DevOps practices by containerizing full-stack projects with Docker and automating their deployment to solve real-world problems. (L5 – Evaluating)					
Detailed Syllabus					
Unit	Description				Duration
1.	Front End Development Introduction of HTML, Advance HTML, CSS, Flexbox, Grid System, Bootstrap or Tailwind Framework, JavaScript, Arrow Function, Promises, Async & Await, ES6+(ECMAScript).				05
2.	Back End Development Essential Python Concepts, OOPs, Networking with Python, File I/O, Django Web Framework, Models, Views, Templating, Authentication and Authorization, Sessions, Database Connectivity, APIs development with Django REST Framework, Machine Learning model integrations				05
3.	Databases and Deployment SQL (Postgres) and NoSQL (MongoDB) Database, AWS S3, Digital Ocean, Infrastructure as a Service (IaaS) platform for software developers, creating droplet.				09



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4.	Reactjs JSX, Context API, Reducer, Provider, Consumer, Hooks, States, Props, API handling, Backend Connectivity	05
5.	Project and DevOps Docker CLI Command, Integrate Django with Docker, Image, Docker Deployment.	06
	Total	30

Text Books

1. Chris Northwood, *The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer*, 1st Edition, Apress publisher, 2018.
2. Fabrizio Romano, Gaston C. Hillar, Arun Ravindran, *Learn Web Development with Python*, 1st Edition, Packt Publishing Limited, 2018.

Reference Books

1. Arun Ravindran, Samuel Dauzon, Aidas Bendoraitis, *Django: Web Development with Python*, 1st Edition, Ingram Publisher, 2017.
2. Aidas Bendoraitis, Jake Kronika, *Django 3 Web Development Cookbook: Actionable solutions to common problems in Python web development*, 4th Edition, Packt Publishers, 2020.
3. Nader Dabit, *Full Stack Serverless: Modern Application Development with React, AWS, and GraphQL*, 1st Edition, Shroff/O'Reilly publisher, 2020.

Laboratory Work

8 to 10 programming exercises (and a practicum) based on the syllabus.



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Program: B Tech, MBA Tech & B Tech Integrated Data Science, B Tech CSE (DS)				Semester: V/ VI/ IX	
Course : Applied Artificial Intelligence				Code: 702DB0C019	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
3	2	0	4	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Python Programming					
Course Objective The aim of the course is to provide accessible coverage of the key areas of AI including game playing, expert systems, natural language understanding, vision, robotics, agents and modelling cognition. It will provide information on various AI based algorithms and methodologies used to solve problems with an AI based approach.					
Course Outcomes After completion of the course, students will be able to - 1. Explain the theory behind graph search algorithms, classification, optimization, reinforcement learning, and other topics in artificial intelligence and machine learning 2. Solve the uncertainty & optimization problems using the AI methods. (L3-Applying) 3. Apply knowledge representation and inference techniques to reason under logical and probabilistic frameworks. (L4- Analyzing) 4. Analyze the performance and trade-offs of AI algorithms in solving real-world problems. (L2- Understanding, L3-Applying) 5. Evaluate and compare different learning algorithms to select the appropriate method for a given task. (K5- Evaluating) 6. Design intelligent systems of their own (K6 - Creating)					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Artificial Intelligence Intelligent agents, State, Actions, Transition model, Path cost, Evaluation functions				03
2.	AI Search Problems Uninformed Search (DFS, BFS), Informed Search (Heuristic Strategies – GBFS, A*), Adversarial Search (Minimax, Alpha-Beta Pruning)				04
3.	AI Knowledge Problems Inference Algorithms, Model Checking, Knowledge base Knowledge Engineering, Inference by Resolution, First – Order Logic, Second order logic, Sentence, Entailment, Conjunctive normal form, Inference				06



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4.	Uncertainty Problems Unconditional probability, Conditional Probability, Joint Probability, Baye's Rules & Bayesian Kings, Inference by enumeration & approximation, Sampling & Uncertainty over time, Likelihood weighting, Markov assumption, chain and models	06
5.	AI Optimization Problems Local Search Hill climbing, steepest-ascent, stochastic problems, first-choice, random-restart, Local Search Algorithms, local beam search, Linear Programming Algorithms, Arc Consistency, Backtracking search	08
6.	AI Based Learning Classification, Regression, k-fold cross-validation, Unsupervised learning, Regularization, Reinforcement Learning, Q Based Learning, epsilon-greedy, k-means clustering	09
7.	Fundamentals of Neural Networks Artificial neural network, Multilayer neural network, Deep neural network, Recurrent neural network, Convolutional neural network, Gradient descent algorithm, Backpropagation algorithm	09
	Total	45
Text Books 1. Stuart Russell, Peter Norvig, <i>Artificial Intelligence: A Modern Approach</i>, 4th Edition, Pearson, 2020. 2. Kevin Knight, Elaine Rich, Shivashankar B. Nair, <i>Artificial Intelligence</i>, 3rd Edition, McGraw Hill Education, 2017.		
Reference Books 1. Thomas H. Davenport, Erik Brynjolfsson, Andrew McAfee, H. James Wilson, <i>Artificial Intelligence: The Insights You Need from Harvard Business Review (HBR Insights Series)</i> Paperback – Illustrated Edition, Harvard Business Review Press, 2019. 2. Richard S. Sutton, Andrew G. Barto, Francis Bach, <i>Reinforcement Learning – An Introduction</i>, 1st Edition, MIT press, 2018. 3. Max Tegmark, <i>Life 3.0: Being Human in the Age of Artificial Intelligence</i>, Penguin Publisher, 2018.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus.		



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Program: B Tech, MBA Tech & B Tech Integrated Data Science				Semester: V/ IX	
Course: Foundations of Machine Learning				Code: 702DB0C020	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
3	2	0	4	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Statistical Structures in Data and Inference					
Course Objective The aim of the course is to focus on the basic principles of machine learning and use the concepts for developing models. The course will emphasize on topics from matrix algebra, calculus, optimization, and probability theory those are having strong linkage with machine learning					
Course Outcomes After completion of the course, students will be able to - 1. Recall and define key concepts of machine learning, including types of learning, model evaluation metrics, and essential terminology. <i>(L1- Remembering)</i> 2. Explain the fundamental differences between supervised vs. unsupervised learning, classification vs. regression, and concepts like bias-variance. <i>(L2-Understanding)</i> 3. Apply foundational machine learning techniques to both supervised and unsupervised learning problems, including regression, classification, and dimensionality reduction. <i>(L3- Applying)</i> 4. Assess and interpret model performance using key evaluation metrics, understand the bias-variance trade-off, and distinguish quality of fit in various settings. <i>(L5 – Evaluating)</i> 5. Understand and apply fundamental concepts of optimization, including gradients, Jacobians, Hessians, gradient descent, and constrained optimization, with applications to training neural networks and building feedforward architectures. <i>(L2-Understanding and L3- Applying)</i> 6. Perform and interpret dimensionality reduction techniques, such as Principal Component Analysis (PCA) and Fisher Discriminant Analysis, and apply support vector machines for classification with kernel methods and geometric interpretations. <i>(L4-Analyzing)</i>					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Machine learning The Trade-Off Between Prediction Accuracy and Model Interpretability, Supervised Versus Unsupervised Learning, Regression Versus Classification Problems. Assessing Model Accuracy Measuring the Quality of Fit, The Bias-Variance Trade-Off , The Classification Setting				06
2.	Essential Linear Algebra for Machine Learning Programming Linear Algebra, Analytic Geometry, Vector arithmetic, Norms, L1 and L2 square Norms, matrix and tensors, special types of matrix, transforming vector space				06
3.	Matrix Decomposition with an application to Data Problems Eigen face and Eigen space, Cholesky Decomposition, Eigen Decomposition and Diagonalization, Building Linear Regression (Ordinary Least Squares) with Linear Algebra				06



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4.	Basics of Convexity Differentiations of Univariate and multivariate functions, Partial differentiations and gradients, Gradients of matrix: Jacobian and Hessians matrix, Multivariate chain rule and its applications, Continuous optimization, Optimization using Gradient Descent, Constrained optimization and lagrange Multipliers, Convex optimization, Feedforward Neural networks, Backpropagation Algorithm, Building Simple Artificial Neural Networks	09
5.	Dimensionality reduction Dimensionality Reduction with principal component Analysis, Maximum variance perspective, Projection perspective, Eigenvector computation and low rank approximations, PCA in high Dimensions, Gaussian Discriminant Analysis, Fisher discriminant analysis for Classification and Dimensionality Reduction	10
6.	Classification with Support Vector Machines Separating hyperplanes, Primal support Vector Machine, Dual Support vector machine, Kernel	08
	Total	45

Text Books

1. M.P. Deisenroth, A. A. Faisal, and C.S. Ong, *Mathematics for machine learning*, 1st Edition, Cambridge University Press, 2020.
2. Gilbert Strang, *Linear Algebra and Learning from Data*, 1st Edition, Cambridge University Press, 2019.

Reference Books

1. Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine Learning*, 3rd Edition, Cambridge University Press, 2017.
2. Mehryar Mohri, Afshin Rostamizadeh and Ameet Talwalkar, *Foundations of machine learning*, 2nd Edition, MIT Press, 2021.
3. Strang, Gilbert, *Linear Algebra for Everyone*, 1st Edition, Wellesley-Cambridge Press, 2020.

Laboratory Work

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Program: B Tech, MBA Tech & B Tech Integrated Data Science, B Tech [AI and DS, CSE (DS)]				Semester: V/ VI/ IX/ V-VII	
Course: Predictive Analysis				Code: 702DB0E003	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Nil					
Course Objective It aim of the course is to educate Students for data preparation, building predictive models, assessing and implementing models for decision making					
Course Outcomes After completion of the course, students will be able to – 1. Understand and classify the structure of programming environments, including libraries, diagrams, and data sources within SAS Enterprise Miner. (L2 – Understand) 2. Apply decision trees and regression models to perform predictive analysis and optimize model complexity using SAS tools. (L3 – Apply) 3. Evaluate statistical measures and model fit statistics to assess and design supervised and unsupervised models effectively. (L5 – Evaluate) 4. Analyze and implement advanced predictive models using neural networks and additional modeling tools to solve real-world data problems. (L4 – Analyze) 5. Apply unsupervised learning techniques such as clustering and market basket analysis for pattern discovery in large datasets. (L4 – Analyze)					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Programing Software				01
2.	Accessing & Assaying Prepared Data Creating SAS E-Miner Project, Library & Diagram, Defining a Data Source in E-Miner, Exploring a Data Source in E-Miner				03
3.	Introduction to Predictive Modelling Fundamentals & Tree Based Method, Introduction to Decision Trees, Cultivating Decision Trees in E-Miner, Optimizing the complexity of Decision Tree, Understanding Additional Diagnostic Tools, Autonomous Tree growth options				04
4.	Introduction to Predictive Modelling Regressions, Introduction to Regressions, Selecting Regression inputs in E-Miner, Optimizing Regression Complexity, Interpreting Regression models, Transforming Inputs, Handling Categorical inputs, Polynomial Regressions				05



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5.	Introduction to Predictive Modelling Neural Networks & Other Modelling tools, Introduction to Neural Network, Input Selections in Neural Network, Stopped Training, Other modelling tools	06
6.	Model Assessment Model Fit statistics, Statistical Graphics, Adjusting for Separate Sampling, Profit Matrices	03
7.	Model Implementation Introduction to model implementation methodology, Internally scored datasets, Score code modules	03
8.	Introduction to Pattern Discovery Introduction to unsupervised learning, Cluster Analysis, Market Basket Analysis	03
9.	Miscellaneous Special Topics Ensemble Models, Variable Selection in Ensemble Modelling, Categorical Input Selection, Surrogate Models	02
	Total	30
Text Books 1. Kattamuri S. Sarma PhD, <i>Predictive Modeling with SAS Enterprise Miner: Practical Solutions for Business Applications</i>, 3rd Edition, SAS Institute publication, 2017. 2. Max Kuhn, Kjell Johnson, <i>Applied Predictive Modeling Hardcover</i>, 1st Edition, Springer publisher, 2018.		
Reference Books 1. Randall Matignon , <i>Data Mining Using Sas Enterprise Miner</i>, 1st Edition, Wiley publication, 2012. 2. Shai Shalev-Shwartz and Shai Ben-David, <i>Understanding Machine Learning</i>, 3rd Edition, Cambridge University Press, 2017. 3. Mehryar Mohri, Afshin Rostamizadeh and Ameet Talwalkar, <i>Foundations of machine learning</i>, 2nd Edition, MIT Press, 2021. 4. Strang, Gilbert, <i>Linear Algebra for Everyone</i>, 1st Edition, Wellesley-Cambridge Press, 2020.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus		



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Program: B Tech, MBA Tech & B Tech Integrated Data Science				Semester: V/ IX	
Course: Simulation Modelling and Analysis				Code: 702DB0E004	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Probability theory					
Course Objective To acquaint the students with various modeling and simulation techniques for discrete, dynamic and stochastic systems. It will focus on the important aspects of simulation study, including modelling, simulation languages, validation, and output data analysis.					
Course Outcomes After completion of the course, students will be able to - 1. Understand the fundamentals of system modeling and simulation techniques, including deterministic/stochastic systems, discrete event simulation, continuous simulation, and Monte Carlo methods.(L2 – Understand) 2. Apply discrete event simulation techniques using time-advance mechanisms and event-driven approaches to model systems such as single-server queues.(L3 – Apply) 3. Develop and simulate real-world systems using GPSS by utilizing blocks, transactions, user-defined functions, and logic switches.(L4 – Develop) 4. analyze random numbers and variates using statistical tests (chi-square, runs test), and select appropriate probability distributions for simulation input modeling.(L4 – Analyze) 5. Evaluate simulation results using statistical techniques such as confidence intervals, variance reduction methods, and experimental design to compare system alternatives. (L5 – Evaluate)					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction Systems, models, deterministic and stochastic systems, static and dynamic systems, discrete event simulation, continuous simulation, Monte Carlo simulation.				03
2.	Discrete Event Simulation Time-advance mechanisms, event modeling of discrete dynamic systems, event graphs, process oriented and event oriented approaches, single-server single queue model.				03
3.	GPSS Program model, entities and transactions, blocks in GPSS, user defined functions, SNA, logic switches, save locations, user chains, tabulation of result, programming examples.				05
4.	Random Number Generation Congruence generators, long period generators, statistical quality measures of generators, uniformity and independence testing, chi-square and other hypotheses testing, runs testing.				04



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5.	Random Variate Generation Random variable, probability density and distribution functions, Location, scale and shape parameters, discrete and continuous probability distributions; Inverse transform method, composition and acceptance-rejection methods, efficiency and quality measures of generators; Input Modelling, selection of distribution for a random source, fitting distributions to data, constructing empirical distributions from data	05
6.	Random Processes and Queuing Models random process, discrete/continuous time processes, Markovian property, and Markov chain.	05
7.	Simulation Output Data Analysis for a Single System statistical analyses for transient systems and systems in statistical equilibrium. Comparing alternative system configurations; confidence intervals, ranking and selection. Variance reduction techniques. Experimental design, sensitivity analysis and simulation-based optimization.	05
	Total	30
Text Books 1. M. Law and W. D. Kelton, <i>Simulation Modeling and Analysis</i>, 5th Edition, McGraw Hill, 2021. 2. Giuseppe Ciaburro, <i>Hands-On Simulation Modeling with Python</i>, 1st Edition, Packt Publishing Limited, 2020.		
Reference Books 1. J. Banks, J. S. Carson, B. L. Nelson and D. M. Nicol, <i>Discrete Event System Simulation</i>, 6th Edition, Pearson Education, 2013. 2. Sheldon M. Ross, <i>Simulation</i>, 5th Edition, Academic Press, 2012. 3. David T Sturrock, W David Kelton, Jeffrey S Smith, <i>Simio and Simulation: Modeling, Analysis, Applications</i>, 5th Edition, Createspace Publishing, 2018.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus		



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Program: B Tech, MBA Tech & B Tech Integrated Data Science				Semester: V/ IX	
Course: Internet of Things Analytics				Code: 702DB0E005	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Prerequisite: Programming for problem solving					
Course Objective The aim of the course is to educate students about the fundamentals of Internet of Things (IoT) and its building blocks along with their characteristics. It will provide students with understanding of protocols and standards designed for IoT and the current research on it. It will help students understand the recent application domains of IoT in everyday life. It will provide information on associated technologies like cloud computing in the domain of IoT					
Course Outcomes After completion of the course, students will be able to - 1. Describe the fundamentals of IoT architectures, including M2M technologies, SMAC stack, and communication protocols essential for IoT environments. L2 – Understand (Linked to Units 1 & 2) 2. Demonstrate the appropriate use of IoT reference models, networking architectures, and protocols such as MQTT, CoAP, and 6LoWPAN in IoT systems. L3 – Apply (Linked to Units 2 & 3) 3. Connect IoT devices to cloud platforms using APIs, SDKs, and messaging protocols to enable scalable and data-driven IoT applications. L3 – Apply (Linked to Unit 4) 4. Design and develop IoT-based applications using Raspberry Pi and GPIO programming for real-time monitoring and control. L4 – Analyze (Linked to Unit 5) 5. Analyze real-world IoT use cases and propose innovative ideas through case studies, research literature, and patent reviews for digital transformation. L4 – Analyze (Linked to Unit 6)					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to IoT Basics of computer networks, IoT-An Architectural Overview- Building an architecture, Main design principles, and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals. Machine-to-machine wireless communication technologies for the Internet of Things Taxonomy, comparison, and open issues. SMAC (Social, Mobile, Analytics, Cloud) Stack - SMAC – Foundational Technology Stack, SMAC – The Basic Enablers.				07
2.	IoT Fundamentals IoT reference Model - IoT Reference Architecture Introduction, Functional View, Information View, Deployment and Operational View, Other Relevant architectural views, Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service (XaaS), M2M and IoT Analytics, Knowledge Management				06



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3.	IoT Protocols Networking Architectures: Star, Mesh, Tree, Networking Protocols: TCP/IP, 6LowPan, RPL, Thread, IoT Devices Application-Level Protocols: MQTT, CoAP, REST	04
4.	IoT and Cloud IoT Devices and Cloud access, Cloud components, Cloud SDK, Analytics API, Cloud API, Cloud registry, MQTT, Cloud protocols for IoT.	04
5.	IoT with Raspberry Pi Raspberry Pi, Programming with GPIO, GPIO library, Basic recipes, Advanced recipes, API Output devices, Advanced recipes, General Purpose Input Output Pins, Network and Raspberry Pi, Network Sockets, Sockets and Server, Network Libraries.	07
6.	Case study - Patents and Research Smart Cities, Smart Home Industrial Control, Smart Social Networks, Big Data Analytics. Reviewing literature from peer-reviewed journals. Proposing patent on IoT application.	02
	Total	30

Text Books

1. Vlasios Tsiatsis, Stamatis Karnouskos, Jan Höller, David Boyle and Catherine Mulligan, *Internet of Things: Technologies and Applications for a New Age of Intelligence Book*, 2nd Edition, Academic Press 2018.
2. Andrew Minter, *Analytics for the Internet of Things (IoT)*, 1st Edition, Packt Publishing Ltd, 2017.

Reference Books

1. Rajkumar Buyya and Amir Vahid Dastjerdi, *Internet of Things: Principles and Paradigms*, 1st Edition, Morgan Kaufmann, 2016.
2. Olof Liberg, Marten Sundberg, Y.-P. Eric Wang, Johan Bergman and Joachim Sachs, *Cellular Internet of Things: Technologies, Standards, and Performance*, 2nd Edition, eBook, 2016.
3. Shancang Li and Li Da Xu, *Securing the Internet of Things*, 1st Edition, Syngress Publisher, 2017.
4. Harry G. Perros, *an Introduction to IoT Analytics*, 1st Edition, CRC Press, 2021.

Laboratory Work

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Program: B Tech, MBA Tech & B Tech Integrated Data Science, B Tech CSE (DS)				Semester: V/ IX	
Course : Data Driven Program Analysis				Code: 702DB0E006	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Programming for problem solving					
Course Objective The aim of the course is to provide an introduction to Data mining and concepts of Business Intelligence. It also outlines data mining as a step in KDD and further provides information on classification methods and frequent pattern mining					
Course Objectives After completion of the course, students will be able to – 1. Describe the key concepts, processes, and advantages of data mining and its applications in real-world business scenarios. 2. Explain the architecture, components, and development phases of Business Intelligence systems and Decision Support Systems. 3. Illustrate the Knowledge Discovery in Database (KDD) process and differentiate between data matrix and dissimilarity matrix with respect to data mining functionalities. 4. Construct and apply basic classification models such as ID3 and Naïve Bayes to solve simple prediction and categorization problems. 5. Analyze association rules using support, confidence, and frequent itemset mining techniques (e.g., Apriori algorithm) for market basket analysis and recommendation systems.					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Data Mining Definition, data mining process: Business understanding, Data understanding, Data preparation, Modeling, Evaluation, Deployment				02
2.	Data Mining advantages and applications Data mining use cases and examples				04
3.	Fundamentals of Business Intelligence Business Intelligence Architecture, components of business intelligent system, decision support system; its features; structure and extended structure, phases in development of business intelligence system				04
4.	Knowledge Discovery in Database Steps in KDD process, Architecture of a data mining system, data mining functionalities, data matrix vs. dissimilarity matrix				04



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5.	Basics of Classification Model construction, Model usage, examples of ID3, Naïve Bayes classifier examples	08
6.	Market Basket Analysis Support, confidence, association rules, Frequent itemset mining using Apriori algorithm examples	08
	Total	30
Text Books 1. E. Frank, I. H. Witten and M. Hall, <i>Data Mining: Practical Machine Learning Tools and Techniques</i>, 3rd Edition, Elsevier 2017. 2. Alex Berson and Stephen Smith, <i>DATA WAREHOUSING, DATA MINING, & OLAP</i>, 1st Edition, McGraw Hill Education, 2017.		
Reference Books 1. Han, J., Kamber, M., & Pei, J., <i>Data mining: Concepts and techniques</i>, 3rd Edition, Elsevier, 2012. 2. M. J. Zaki and W. Meira, <i>Data Mining and Analysis: Fundamental Concepts and Algorithms</i>, 1st Edition, Cambridge University Press, 2014. 3. Jesus Salcedo, <i>Machine Learning for Data Mining</i>, Ebook 1st Edition, Packt Publishing Limited, 2019.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus.		



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Program: B Tech/ MBA Tech (Data Science and Artificial Intelligence), B Tech [AI and ML, AI and DS, CSE (DS)]/ B Tech Integrated Data Science				Semester : V/VII/ VIII/ IX	
Course: Applied Time Series Analysis				Code : 702DB0E007	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Statistical Structures in Data and Inference					
Course Objective This course provides an introduction to time series analysis. It will educate students on how to choose an appropriate forecasting method in a particular environment and then apply it using necessary computations. It aims to equip students with various forecasting techniques and knowledge on modern statistical methods for analyzing time series data. It will also provide directions to Improve forecasting using better statistical models based on statistical analysis					
Course Outcomes After completion of the course, students will be able to – 1. Explain the fundamental components of time series data, such as trend, seasonality, and autocorrelation, and visualize them using various plotting techniques. 2. Apply time series regression models, exponential smoothing, ARMA, ARIMA, and SARIMA models to forecast univariate and seasonal data. 3. Analyze stationary and non-stationary time series processes using ACF, PACF, unit root tests, and diagnostics to identify appropriate forecasting models. 4. Evaluate forecasting performance of hierarchical and grouped time series models using reconciliation methods and compare with alternative approaches. 5. Design advanced forecasting solutions using complex seasonality, the Prophet model, VARs, bootstrapping, and bagging on real-world datasets using statistical software.					
Detailed Syllabus					
Unit	Description	Duration			
1.	Examples of Time Series The basic steps in a forecasting task Time Series Graph Seasonal plots, Seasonal subseries plots, Lag plots, Autocorrelation and time series decomposition, Stationary models and autocorrelation function Estimation and elimination of trend and seasonal components	03			
2.	Time Series Regression Models and Exponential Smoothing	02			
3.	Stationary Process and ARMA Models Basic properties and linear processes, Introduction to ARMA models, properties of sample mean and autocorrelation function, Forecasting stationary time series ARMA(p, q) processes, ACF and PACF, Forecasting of ARMA processes.	05			



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4.	Modeling and Forecasting with ARMA Processes Preliminary estimation, Maximum likelihood estimation, Diagnostics, Forecasting Order selection.	05
5.	Nonstationary and Seasonal Time Series Models ARIMA models, Identification techniques, Unit roots in time series, Forecasting ARIMA models, Seasonal ARIMA models, Regression with ARMA errors	03
6.	SARIMA - Seasonal Autoregressive Integrated Moving Average	02
7.	Forecasting Hierarchical and Grouped Time Series Hierarchical and grouped time series, Single level approaches, Forecast reconciliation	03
8.	Advanced Forecasting Methods Complex seasonality, Prophet model, Vector auto regressions, Bootstrapping and bagging	07
	Total	30
Text Books - Galit Shmueli and Kenneth C. Lichtendahl , <i>Practical Time Series Forecasting with R: A Hands-On Guide</i>, 2nd Edition, Axelrod Schnall Publishers, 2016. - Damodar Gujarati, Dawn Porter, Sangeetha Gunasekar, <i>Basic Econometrics</i>, 5th Edition, McGraw Hill Education, 2017.		
Reference Books - Jenkins and Reinsel, <i>Time Series Analysis: Forecasting and Control</i>, 5th Edition, Prentice Hall, 2015. - Robert Shumway, David Stoffer, <i>Time Series Analysis and Its Applications with R examples</i>, 4th Edition, Springer, 2016. - Rami Krispin , <i>Hands-On Time Series Analysis with R: Perform time series analysis and forecasting using R</i>, 1st Edition, Packt Publishing Limited, 2019.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus.		



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Program: B Tech, MBA Tech & B Tech Integrated Data Science, B Tech CSE (DS)				Semester: V/ IX/ VI	
Course: Virtualization and Cloud Computing				Code: 702DB0E008	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Database management system, Information security and privacy					
Course Objective To educate students on building cloud infrastructure based on a cloud computing reference model. It will focus on technologies, components, processes, and mechanisms of cloud infrastructure. It will educate students on various algorithms involved in better implementing the cloud-based systems. This course will enable students to take the EMC Cloud Infrastructure and Services Associate Certification (EMCCIS).					
Course Outcomes After completion of the course, students will be able to - 1. Describe the foundational concepts of computer networks and cloud computing, including network architectures, models, protocols, and cloud service/deployment models. (L1 - Remember) 2. Demonstrate the use of virtualization technologies and container tools such as hypervisors, Docker, and Kubernetes for cloud infrastructure. (L3 - Apply) 3. Implement and configure core AWS services including EC2, IAM, and S3 for secure and scalable cloud deployments. (L3 - Apply) 4. Analyze resource management techniques such as load balancing, auto scaling, and provisioning in distributed cloud systems. (L4 - Analyze) 5. Develop and deploy cloud-based applications using advanced AWS tools like Lambda, CloudFormation, SageMaker, and monitoring services. (L6 - Create)					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to computer networks, Network Models, OSI Models and TCP/IP , Internet, Protocols, Network Applications				03
2.	Introduction to Cloud Distributed Computing and Enabling Technologies, Cloud Fundamentals: Cloud Definition, Evolution, Architecture, Applications, Essential characteristics of cloud computing, cloud deployment models, and cloud service models. Definition, Types of Virtualization, Hardware Virtualization, Full and Para Virtualization, Hypervisors, Hardware-assisted virtualization, operating system level virtualization, application virtualization, Containers, Docker and Kubernetes				06



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3.	Introduction to AWS Various Platforms for Cloud Computing, Why AWS, Creating AWS Account, Basic Concept. Identity and Access Management Overview, IAM Use cases and service details, IAM Users-Groups-Policies, MFA, AWS CLI setup, Roles, Security Tools	03
4.	Elastic Compute Cloud(EC2) Basics Overview, Security Groups, SSH, EC2 Instance Connect. Implementation Study of Cloud computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open source tools.	04
5.	Elastic Load Balancing (ELB) and Auto Scaling Groups(ASG) High Availability, Scalability, Elasticity, Agility, Overview Applications for ELB and ALB, ASG along with hands-on Resource Management and Load Balancing Distributed Management of Virtual Infrastructures, Server consolidation, Dynamic provisioning and resource management, Resource Optimization, Scheduling, Load Balancing, various load balancing techniques.	04
6.	S3, Databases and Analytics Security Bucket Policy, Versioning, Website, Replication, Snowball, Snowball Edge and Snowmobile Overview Databases RDS and Aurora, Overview, ElastiCache, DynamoDb, RedShift Map Reduce and its extensions to Cloud Computing, HDFS, and GFS.	04
7.	Other Compute Services Lambda, ECS, Light sail, Cloud Formation, Route 53, Cloud Front, SNS and SQS, Cloud Watch, Cloud Trail, Amazon Elastic Beanstalk, Cognito, Sage Maker, Amazon SES, Amazon Polly, Transcribe, Customer carbon footprint tool, Amazon sustainability data initiative .	06
	Total	30
Text Books 1. Ian Foster and Dennis B. Gannon, <i>Cloud Computing for Science and Engineering</i> , 1 st Edition, MIT Press, 2017. 2. Kevin L. Jackson, Scott Goessling, <i>Architecting Cloud Computing</i> , 1 st Edition, Packt Publishing Limited, 2018.		



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Reference Books

1. Anthony T. Velte, *Cloud Computing: A Practical approach*, 1st Edition, Tata Mcgraw Hill Education Private Limited, 2017.
2. Marinescu D C, *Cloud Computing Theory and Practice*, 2nd Edition, Elsevier, 2017.
3. Andreas Wittig and Michael Wittig, *Amazon Web Services in Action*, manning publisher, 2018.
4. William Stallings, *Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud*, 1st Edition, Addison-Wesley Professional, Pearson, 2015.

Laboratory Work

8 to 10 programming exercises (and a practicum) based on the syllabus



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Program: B Tech (Civil, Mechanical & EXTC) except MBA Tech (all programs) and B Tech (IT, AI, DS, CE, AIDS, AIML, Cyber Security, Mechatronics, CSBS, CSE (DS))/ BTI DS				Semester : V/VI	
Course: Data Analytics				Code : 702DB0I001	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
3	0	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Nil					
Course Objective Data analysis is the method of examining, cleansing, and modeling data with the objective of determining useful information for effective decision-making and operations. It aims to provide knowledge to students to perform analysis using different analytical tools. It focuses on the need, importance and applications of analytics in different domains					
Course Outcomes After completion of the course, students will be able to – 1. Apply concepts of data manipulation and wrangling 2. Analyze and work on raw data using analytical software 3. Create models using programming, descriptive statistics, and data architecture strategies 4. Interpret analytical results to support data-driven decision-making in various business domains. 5. Design visual representations of data using charts and statistical summaries to communicate insights effectively.					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction to Data Analytics Overview of the Data Analyst Ecosystem				03
2.	Understanding various data types Understanding Different Types of File Formats , Languages for Data Professionals, Overview of Data Repositories				03
3.	Identifying Data for Analysis Different Data Sources, Gather and Import Data , Data Wrangling				05
4.	Tools for Data Wrangling Data Cleaning, Data Preparation and Reliability, Data transformation, Data merging				07
5.	Date manipulation, Data Charts				04
6.	Overview of Statistical Analysis Descriptive Statistics, Grouping Data in Data Analytics software, Analysis of Variance (ANOVA), Correlation Analysis				08
7.	Introduction to Model Development Simple Linear Regression, Multiple Linear Regression, prediction and decision making				07



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8.	Applications Data Analytics for strategy formation, marketing and customer analytics, digital social media, operations and supply chain	08
	Total	45
Text Books 1. Garrett Grolmund and Hadley Wickham, <i>R for Data Science</i> , 1 st Edition, Oreilly, 2017. 2. Anil Maheshwari, <i>Data Analytics</i> , 1 st Edition, McGraw Hill Education, 2017.		
Reference Books 1. Richard Levin, David S Rubin, Sanjay Rastogi and Massod Husain, <i>Statistics For Management</i> , 8 th Edition, Pearson, 2017. 2. Bharati Motwani, <i>Data Analytics with R</i> , 1 st Edition, Wiley publisher, 2019. 3. Bharati Motwani, <i>Data Analytics using Python</i> , 1 st Edition, Wiley publisher, 2019.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus		



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Program : B Tech programs and BTI DS except B Tech CSBS and CSE (DS) and MBA Tech all programs				Semester: V/ VI/ IX / X	
Course: Basic Decision making with Spreadsheet				Code: 702DB0I002	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks - 50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Nil					
Course Objective The aim of the course is to provides comprehensive instruction using spreadsheet software as a productive decision making tool. Further, it will enable students to apply spreadsheet capability, analyze various models influencing variables from period to period and evaluate data patterns to solve possible outcomes of an uncertain event.					
Course Outcomes After completion of the course, students will be able to - 1. Describe the fundamental capabilities, notations, and functions of spreadsheets for creating data models and solving mathematical expressions. 2. Apply probability distributions, decision trees, and regression techniques in spreadsheets to handle uncertainty and randomness in business decision-making. 3. Develop spreadsheet-based decision models including cash-flow, what-if, and sensitivity analyses for evaluating business scenarios and performance. 4. Solve optimization problems such as linear programming, transportation, assignment, and multi-objective problems using Excel Solver and other spreadsheet tools. 5. Construct simulation and forecasting models using Monte Carlo simulation, inventory and queuing simulations, and time series analysis (e.g., exponential smoothing, seasonality, trend).					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction Introduction to spreadsheets; historical development; basic capabilities of spreadsheets and their usage for creating models; types of data used in spreadsheets; spreadsheet notations for mathematical operations; common built-in formulas and functions; conditional expressions; relative and absolute references.				05
2.	Addressing Uncertainty and Probability Random variables and probability distributions, Changes in discrete and continuous time, Power, exponential, and log functions, Probability trees and decision trees, Correlation and Regression., Simulation and Optimization				05



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3.	Model building Designing spreadsheets reflecting assumptions; decision variables; and outcomes, creating basic cash-flow models; revaluating small business opportunities; incorporating what-if analysis; identifying key variables using sensitivity analysis; linear programming models and deterministic models.	06
4.	Optimization with Spreadsheets using Solver Linear programming, sensitivity analysis, transportation and assignment problems, network optimization problems, integer and nonlinear programming, multi-objective optimization, applications of optimization in different areas.	05
5.	Simulation and Optimization Use of spreadsheets to implement Monte Carlo simulations and linear programs for optimization; model uncertainty and risk in spreadsheets; and use of Excel's solver What-if analysis, Inventory simulation, Waiting Line simulation, Simulation with two Quality Inspectors, other simulation issues.	06
6.	Time Series Analysis Forecasting Time Series Patterns, Moving Averages and Exponential Smoothing, Linear Trend Projection, Seasonality.	03
	Total	30

Text Books

1. David Anderson, Dennis J. Sweeney, Thomas Arthur Williams, *Introduction to Management Science: Quantitative Approaches to Decision Making*, 15th Edition, CENGAGE Learning Custom Publishing, 2018.
2. Cliff Ragsdale, *Spreadsheet Modeling and Decision Analysis: A Introduction to Business Analytics*, 8th Edition, Cengage Learning India Pvt. Ltd., 2018.

Reference Books

1. Elliot Bendoly Excel Basics to Blackbelt: An Accelerated Guide to Decision Support Designs, 2nd Edition, Cambridge University Press, 2013.
2. Cliff Ragsdale, *Spreadsheet Modeling and Decision Analysis: A Practical Introduction to Business Analytics*, 8th Edition, South-Western College Publishing, 2014.
3. Hillier and Hillier, *Introduction to Management Science: A Modeling and Case Studies Approach with Spreadsheets*, 5th Edition, McGraw-Hill/Irwin, 2019.

Laboratory Work

8 to 10 programming exercises (and a practicum) based on the syllabus



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Program: B Tech ,MBA Tech programs and BTI DS except B Tech CSBS and CSE (DS)				Semester: V/ VI/ IX/ X	
Course: Algorithmic Trading				Code: 702DB0I003	
Teaching Scheme				Evaluation Scheme	
Lecture (Hours per week)	Practical (Hours per week)	Tutorial (Hours per week)	Credit	Internal Continuous Assessment (ICA) (Marks -50)	Term End Examinations (TEE) (Marks - 100)
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50
Pre-requisite: Nil					
Course Objective The aim of the course is to provide Students with basic consortium related to market trading. It also provides information on development of real-time strategies. It focuses on creating trading engines that will be supported by advance data analytics					
Course Outcomes After completion of the course, students will be able to - 1. Describe the fundamental concepts of algorithmic trading including market structure, order types, brokers, dealers, and various market participants. 2. Analyze different chart patterns and technical indicators (e.g., SMA, EMA, RSI, MACD, Bollinger Bands) to interpret market trends and price movements. 3. Apply suitable trading strategies such as scalping, arbitrage, mean-reverting, and sentiment-based strategies in different trading environments (e.g., equities, derivatives, forex). 4. Evaluate the performance and effectiveness of various algorithmic trading strategies using backtesting methods while identifying potential pitfalls. 5. Design a simple rule-based or indicator-based trading strategy and implement it in a simulated or real-time trading environment using technical and algorithmic tools.					
Detailed Syllabus					
Unit	Description				Duration
1.	Introduction Basics of Algorithmic Trading Different types of order, order properties, market structure, Brokers, dealers, Bid/ Ask spreads, Block trades, Arbitrage, Buy side trades				05
2.	Technical Trading Basic Philosophy of Technical Approach for Investment, Dow theory, Chart construction, Basic concept of trends, Technical Analysis Applied to Different Trading Mediums and Time Dimensions Pattern study Major reversal pattern: Head and shoulders, Double top and double bottom Triple top and triple bottom Continues pattern Triangles, Flags and pennants, Wedge and Rectangle.				10



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	Major indicators & oscillators Simple moving average, Exponential moving average, Relative strength index, Moving average convergence/divergence(MACD), Bollinger Bands, and Stochastic	
3.	Trading Strategies Implementation of Scalping, Scaling, Advance Jobbing & Trend Jobbing in Live Market Environment Triangular Arbitrage Strategies for Forex market Mean Reverting Strategies like Pair Trading in derivatives market Basket Strategy (Index-Index, Index-Stocks) Statistical Arbitrage Strategies Pre & Post Result based Trading Strategies using Sentiment Indicators Overview on High Frequency Trading	08
4.	Back testing Implementation of A Simple Backtester Trading Strategy, Pitfalls, Components,	07
	Total	30
Text Books 1. Yves Hilpisch, <i>Python for Algorithmic Trading: From Idea to Cloud Deployment</i>, 1st Edition, O'Reilly Media, 2020. 2. Stefan Jansen, <i>Hands-On Machine Learning for Algorithmic Trading: Design and implement investment strategies based on smart algorithms that learn from data using Python Paperback – Import</i>, 1st Edition, Packt Publishing Limited, 2018.		
Reference Books 1. Murphy, J., <i>Technical Analysis of the Financial Markets</i>, 1st Edition, Prentice Hall Press, 2020. 2. Nison, S., <i>Japanese Candlestick Charting Techniques</i>, 2nd Edition, Penguin USA, 2001. 3. Ernest P. Chan, <i>Machine Trading: Deploying Computer Algorithms to Conquer the Markets (Wiley Trading)</i>, 1st Edition, Wiley, 2016.		
Laboratory Work 8 to 10 programming exercises (and a practicum) based on the syllabus		



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